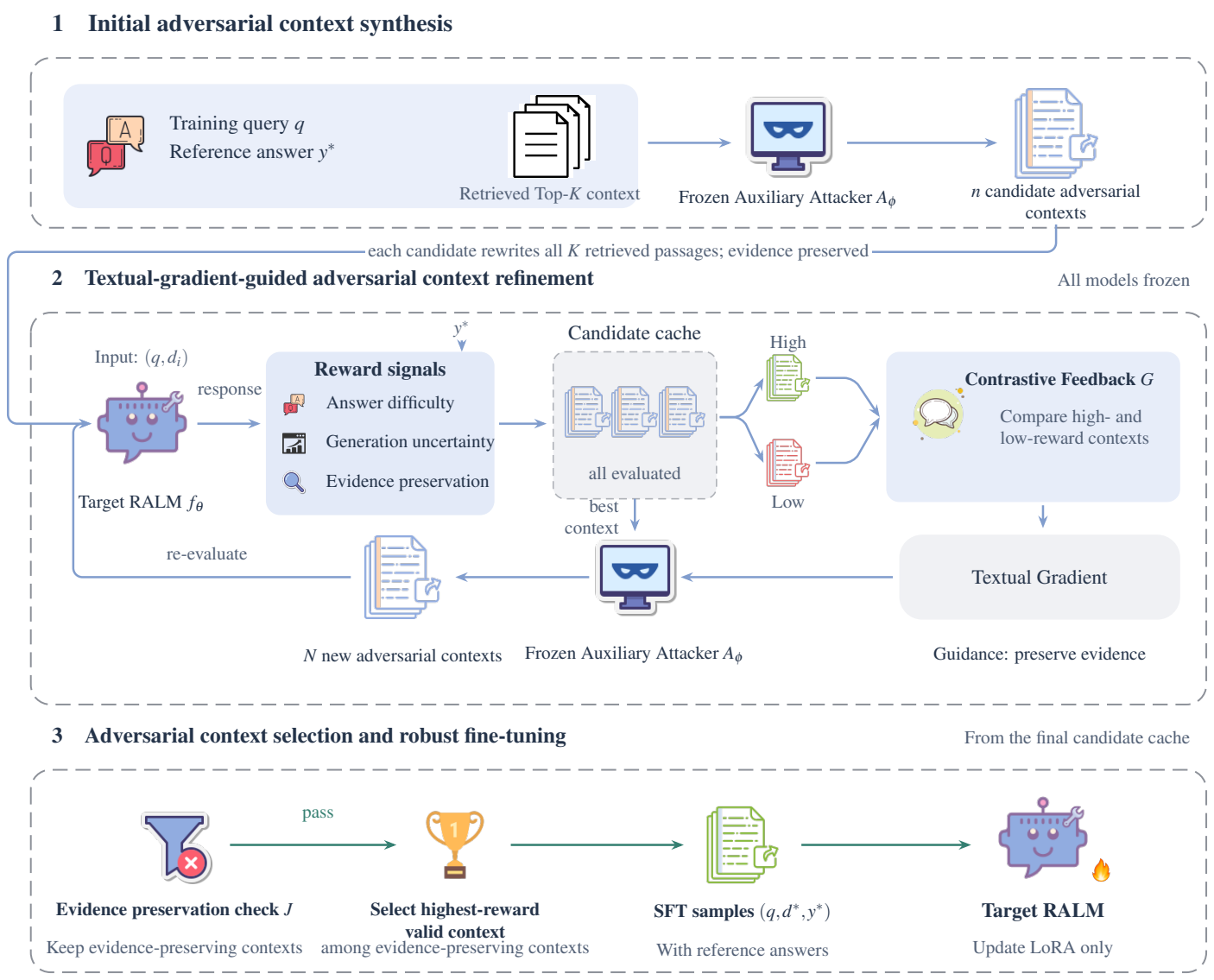

1 Initial adversarial context synthesis
Training query q
Reference answer y^*
Retrieved Top-K context
Frozen Auxiliary Attacker A_ϕ
n candidate adversarial contexts
each candidate rewrites all K retrieved passages; evidence preserved
2 Textual-gradient-guided adversarial context refinement
All models frozen
Input: (q, d_i)
response
Target RALM f_θ
y^*
Reward signals
Answer difficulty
Generation uncertainty
Evidence preservation
Candidate cache
all evaluated
best context
High
Low
Contrastive Feedback G
Compare high- and low-reward contexts
Textual Gradient
Guidance: preserve evidence
Frozen Auxiliary Attacker A_ϕ
N new adversarial contexts
re-evaluate
3 Adversarial context selection and robust fine-tuning
From the final candidate cache
pass
Evidence preservation check J
Keep evidence-preserving contexts
Select highest-reward valid context
among evidence-preserving contexts
SFT samples (q, d^*, y^*)
With reference answers
Target RALM
Update LoRA only